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TCS-401

B. Tech. (CS) (Fourth Semester) End Semester EXAMINATION, 2014

COMPUTER BASED NUMERICAL AND STATISTICAL TECHNIQUES

Time : Three Hours]

[Maximum Marks : 100

Note : (i) This question paper contains two Sections :
Section A (Part I and II) and Section B.

(ii) Answer all questions in Section A (Part I)
briefly in not more than 50 words. Each
question carries 2 marks.

(iii) Answer any *four* questions from Section A
(Part II). Each question carries 6 marks.

(iv) Answer any *four* questions from Section B.
Each question carries 14 marks.

Section—A

Part—I

2 each

1. Attempt all questions briefly in not more than 50
words :

(i) If $\frac{dy}{dx} = -y$, $y(0) = 1$, then by Euler's
method find the value of $y(0.2)$ taking
 $h = 0.1$.

- (ii) If 0.333 is the approximate value of $1/3$, find the absolute and relative errors.
- (iii) The number of intervals required in Simpson's $1/3$ rule and Simpson's $3/8$ rule is a multiple of and respectively.
- (iv) What is the rate of convergence of Regula-Falsi method?
- (v) Write the Newton's forward difference formula for interpolation.
- (vi) Define linear and non-linear regression.
- (vii) If the regression coefficients are 0.8 and 0.2, what would be the value of coefficient of correlations?
- (viii) Write Simpson's $3/8$ rule of Integration.
- (ix) State intermediate value property.
- (x) Define the averaging operator μ .

Part—II

6 each

2. Write short notes on any *four* of the following :

- (a) Find the approximate root of the equation $x \log_{10} x = 1.2$ using Bisection method upto four approximations.
- (b) Evaluate the value of $\log_e 2$ by finding $\int_0^1 \frac{2x}{1+x^2} dx$ using Simpson's $1/3$ rule, by dividing the interval into four equal parts.
- (c) Solve by Gauss elimination method :

$$2x + 4y + z = 3$$

$$3x + 2y - 2z = -2$$

$$x - y + z = 6$$

(d) Find the missing term of the given data :

x	y
1	7
2	*
3	13
4	31
5	37

(e) Using the method of least squares, fit a straight line to the following data :

x	y
1	14
2	27
3	40
4	55
5	68

(f) Find the solution of $\frac{dy}{dx} = xy + y^2$, $y(0) = 1$, at $x = 0.1$ and $x = 0.2$ by using Runge-Kutta fourth order method.

Section—B

14 (7+7) each

Note : Attempt any *four* questions.

3. (a) Apply Gauss forward formula to obtain $f(x)$ at $x = 3.75$ from the table below :

x	y
2.5	24.145
3	22.043
3.5	20.225

4	18.644
4.5	17.262
5	16.047

- (b) If $r_{12} = .75$, $r_{13} = .65$, $r_{23} = .55$, $\sigma_1 = 9$, $\sigma_2 = 7$, $\sigma_3 = 4$. Determine the regression equation of X_2 on X_1 and X_3 .

4. (a) Find the solution of the equation at $x = 0.1$ using Picard's method, correct to three decimal place :

$$\frac{dy}{dx} = 1 + xy, \quad y(0) = 1$$

- (b) Find the least square fit of the form $y = a + bx^2$ of the following data :

x	y
-1	2
0	5
1	3
2	0

5. (a) Given that :

x	y
1.0	0.841
1.1	0.891
1.2	0.932
1.3	0.963

Find $\frac{dy}{dx}$ at $x = 1.0$.

(b) Calculu
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17.262

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data :

y

2

5

3

0

y

0.841

0.891

0.932

0.963

- (b) Calculate the five yearly moving average of acres under tea in India from the following data :

Year	Area (in acres)
1925	672
1926	679
1927	690
1928	702
1929	712
1930	802
1931	807
1932	809

6. (a) Find the number of terms of the exponential series such that their sum gives the value of e^x correct to 6 decimal places at $x = 1$.
(b) Using Lagrange's formula find $f(x)$ corresponding to $x = 10$:

X	Y
5	12
6	13
9	14
11	16

7. (a) From the ahead series of annual data, find the trend line of semi-averages. Also estimate the value for 2009 :

Years	Actual Value
2000	170
2001	231
2002	261
2003	267
2004	278
2005	302
2006	299
2007	298
2008	240

- (b) Draw the $\bar{X}$ and R chart from the following data. Is the process under control ?

Sample No.	$\bar{X}$	R
1	11.0	4
2	10.4	5
3	10.8	9
4	11.2	4
5	11.8	4
6	11.6	7
7	9.6	7
8	9.6	8
9	10.0	8

8. (a) Solve the given system by Gauss-Seidal iteration method :

$$20x + y - 2z = 17; 3x + 20y - z = 18;$$

$$2x - 3y + 20z = 25$$

- (b) Obtain the regression line of y on x for the following data :

x	y
1	1
3	2
4	4
6	4
8	5
9	7
11	8
14	9

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